This article has been accepted for publication in IEEE Transactions on Power Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPWRS.2026.3670270

Asynchronous Distributed Scheduling of Active Distribution Network and Thermostatically Controlled Loads with MPC-based Aggregation

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***Abstract*—The efficient aggregation, reliable scheduling and privacy protection can improve dispatching capacities of thermostatically controlled loads (TCLs). This paper proposes a statistically feasible chance-constrained programming (CCP) model for the active distribution network (ADN) with multiple TCL aggregators in a distributed manner. The distribution system operator (DSO) minimizes the operation cost of ADN under net load uncertainties. We construct the data-driven uncertainty set from samples based on an optimal shape approximation method and reformulate joint CCP into the tractable robust optimization. For the TCL dispatching, a model predictive control (MPC) based aggregation method is proposed to build TCL aggregators and optimize the operation of TCLs under ambient uncertainties. Furthermore, an asynchronous distributed algorithm is designed to solve the coordination problem between the ADN and TCLs as well as preserving participants' privacy. Case studies demonstrate the effectiveness and reliability of the proposed model and algorithm for improving the economy of system participants under various uncertainties compared with other methods.**

***Index Terms*—Thermostatically controlled load, active distribution network, joint chance-constrained programming, model predictive control, asynchronous distributed algorithm.**

NOMENCLATURE

A. Sets and Parameters

R, R^T	The sets of real numbers and real T-vectors.
$(\mathcal{N}, \mathcal{E})$	The sets of nodes and edges in ADNs.
v_0	The nominal voltage at the substation.
Ω_G, Ω_{AGG}	The sets of DGs and TCL aggregators.
$\boldsymbol{e}$	The column vector of all ones.
Δt	Day-ahead sampling rate of TCLs (1 hour)
t / T	Index of time period/ dispatching horizon.
a, b, c_i^G	Cost coefficients of i-th distributed generator (DGs).
$p_t^{P,R}$	The day-ahead energy and reserve prices of the wholesale market.
$V_{\max/\min}$	The boundaries of nodal voltage in ADNs.
$I_{\max}$	Upper bound of the branch current
$P_{i,\max/\min}^G$	The active power limits of i-th DGs.
$Q_{i,\max/\min}^G$	The reactive power limits of i-th DGs.

B. Variables

$P(Q)_{i,t}^G$	The (re)active power of i-th DG at time t.
$P(Q)_{i,t}^{net}$	The (re)active power of scheduled net load at i-th node at time t.
$P / Ru / Rd_{i,t}^{AGG}$	The power and up/down-reserve of k-th TCL aggregator at time t.
$\theta_{i,t}, \theta_t^{out}$	Indoor and Outdoor temperature.

This work was supported by the National Key Research and Development Program, Grant No. 2024YFB4206500 and the National Natural Science Foundation of China, Grant No. 52277107 (Corresponding Author: Yinliang Xu).

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I. INTRODUCTION

MASSIVE distributed energy resources (DERs) have been integrated in the distribution networks (DNs), which forms the active DNs (ADNs) [1]. Due to the uncertainties of renewable energy resources (RESs) and loads from both generation and demand sides, the scheduling flexibility of controllable loads and DERs becomes crucial to the stable operation of ADNs. One promising way is the coordinated operation of thermostatically controlled loads (TCLs) via demand response (DR) due to their thermal inertia and fast response characteristics [2].

To achieve reliable scheduling of the ADN and TCLs under external uncertainties, the existing researches mainly employ scenario-based stochastic optimization (SO) [3] and robust optimization (RO) [4]. The SO needs to generate considerable scenarios based on the particular probability distribution and suffers from the heavy computation burden. Besides, its practicality heavily relies on the quantity and quality of the sample set. The RO can enhance the computational efficiency but it focuses on the worst case which leads to an over-conservative solution [5]. Thereby, the chance-constrained programming (CCP) is used to balance the reliability and economy of decisions by allowing a small probability of constraint violation. One approach to solving CCP is to assume that the uncertainties follow a specific probability distribution, such as Gaussian distribution [6]. To avoid the distribution assumption in CCP, the distributionally robust chance constraint (DRCC) optimization becomes a promising

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way since it models random variables following distributions that belongs to a family of underlying distributions, the so-called ambiguity set [7]. In addition, the ambiguity set can be defined by moment constraints [8] or appropriate distance functions (e.g., Wasserstein metric [9]). Ref.[8] introduces a DRCC-based optimal power flow (OPF) model under uncertain wind power. However, the moment-based ambiguity set leads to a conservative solution due to the reformulation by Chebyshev inequality. In contrast, the Wasserstein metric-based method can leverage the information of historical data [9] but it suffers from low computing efficiency when the sample size is large.

The above researches use individual chance constraints (ICCs) to model uncertainties which overlook the cumulative impact of multiple constraints. Thereby, the joint chance constraints (JCCs) are used to ensure system-wide reliability. To solve JCC-based scheduling, Ref.[3] adopts Bonferroni approximation (BA) to decompose JCCs into ICCs with ε / n risk level. An optimized BA algorithm is proposed to reduce conservatism by optimizing risk levels of each ICC [10]. However, the BA-based method exhibits significant conservatism with low risk level. To directly solve JCCs, Ref. [11] proposes a Monte Carlo based sequential convex approximation (SCA) algorithm. A data-driven distributionally robust JCC programming (DRJCCP) method based on Wasserstein ball is proposed in [12]. These approaches mitigate the conservatism compared to BA-based method. However, they might increase the computational time due to introducing sample related decision variables. Recently, the statistically feasible robust optimization is explored in some researches [13]. This method considers RO as a tractable approximation of joint CCP (JCCP) and integrates data into uncertainty set construction with the guarantee of rigorous statistical performance. This idea is further explored in unit commitment (UC) [14] and in integrated energy systems (IES) [15]. These studies form uncertainty sets by self-improving reconstruction method developed in [13]. However, the optimal uncertainty set might be accessible when the JCCs satisfy certain special structure and it can further enhance the economic performance.

Owing to the limited capacities and large quantities of TCLs, the direct dispatching of individual TCLs imposes heavy computational burdens to the distribution system operator (DSO). Therefore, the TCL aggregators play an important role in exploiting the scheduling flexibility and enhancing the practicability of TCLs [16]. TCL aggregators can participate in the electricity market to reduce the energy cost and provide ancillary services [17]. However, it is challenging for TCL aggregators to estimate the aggregated flexibility of TCLs accurately since the exact aggregation operations (e.g., Minkowski sum (M-sum) [18]) are the NP-hard problems. To tackle the above challenges, many approximation methods are explored. Firstly, the direct aggregation by summing up the boundaries of individual DERs with average parameters is used in [16]. The clustering method is further introduced to manage massive DERs based on multi-time-scale scheduling in [19]. However, these two methods may suffer from low accuracy and leads to infeasible solutions when DERs are highly heterogeneous. Therefore, the inner-approximation methods based on geometric techniques are proposed which can avoid constraint violations of individual DERs. In [20], a polytope-based aggregation approach is designed to efficiently calculate the M-sum of TCLs. Furthermore, Zhang *et al.* propose an improved aggregation approach by minimal representation and reconstruction operations to accelerate the computation [21]. However, these polytope-based methods suffer from over-conservative results when the heavy heterogeneity exists among individual TCLs in high-dimensional feasible region. Recently, more generalized inner approximation methods by affine transformations of a given convex polytope are studied (e.g., for electric vehicles in [22] and TCLs in [3]). The affine transformations provide a more efficient way to aggregate DER flexibility. Nevertheless, the flexible capacities of TCLs are strongly influenced by their current states. The multi-step scheduling optimization of one TCL aggregator based on stationary prototype will deteriorate the aggregation accuracy. Thus, a more adaptive method with dynamic updating mechanism should be further explored.

To coordinate the operation between the ADN and TCLs, the centralized optimization is adopted in [23]-[25]. The DER aggregators submit their operation ranges to DSO and DSO directly dispatches them in a centralized way. However, these centralized coordination methods might raise privacy concerns and heavy computational complexity. Thereby, the distributed approaches are utilized in the literature. For example, a distributed control with a two-layer hierarchical structure is proposed for TCLs [26]. A distributed alternating direction method of multipliers (ADMM) scheduling model among DER agents is designed in [27]. Ref. proposes an improved ADMM method for coordinated dispatch of RESs and hydrogen systems in a distributed manner [28]. However, the convergence speed and asynchronous communication might limit the practical implementation of distributed methods.

Inspired by aforementioned issues, this paper proposes an asynchronous distributed scheduling framework for the DSO with multiple TCL aggregators under external uncertainties. Novel uncertainty set construction and aggregation methods are designed to improve the economy and flexible capacities, respectively. The key contributions are summarized as follows.

1) In Section II, we first establish the JCC-based scheduling model for the ADN and reformulate it into the statistically feasible robust optimization with data-driven uncertainty set. Since the solution conservativeness is highly related to the uncertainty set, we propose an optimal shape approximation (OSA) method for uncertainty set construction, yielding better solutions compared to the existing methods in [10]-[15], as verified in the case studies.

2) In Section III, we first formulate the polytope-based aggregation model for TCL aggregators. To improve the aggregated feasible region calculated by the conventional static aggregation methods in [3] and [20]-[22], we extend the static aggregation into dynamic aggregation by embedding model predictive control (MPC), which dynamically updates the prototype of TCL aggregators after TCL scheduling at each step and effectively increases the aggregated feasible region.

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3) In Section IV, we divide the original centralized scheduling problem into the distributed coordination model for DSO and TCL aggregators. In contrast to [23]-[25] which allow DSO directly dispatch DERs in a centralized way, and [24]-[28] which do not consider communication failure, we develop an asynchronous distributed algorithm based on relaxed ADMM (R-ADMM) with adaptive penalty factor, which achieves the privacy protection, higher convergence speed and comparable solution accuracy under communication failure.

Notation: We use boldface letters to denote matrices and vectors with appropriate dimensions. The shorthand $[T] = \{1,\ldots,T\}$ represents the set of all integers up to T.

II. Statistically Feasible Scheduling Model of Active Distribution Networks

In this section, we first state the preliminaries and establish the JCC-based scheduling model of ADN under net load uncertainties. Then, we reformulate it into the statistically feasible scheduling model with RO technique. To mitigate RO's conservativeness, the OSA method is proposed to learn the optimal uncertainty set.

A. Preliminaries

Firstly, in the studied DSO-aggregator coordination architecture, TCL aggregators are allowed to participate and bid for quantities in the local market to minimize their operational costs, while DSO is responsible for clearing the market and ensuring the safe operation of distribution network, which aligns with the power market in China.

It should be noted that this structure is different from that of Australian national electricity market. In Australian national electricity market, the DSO only acts as a neutral network operator to guarantee network constraints and calculate the dynamic operating envelopes (DOEs) for DER aggregators. Then, DER aggregators bid into the power market with these DOEs and the market operator clears them without any functioning DSO [29].

Secondly, TCLs are considered as flexible active power resources and their reactive power is not explicitly dispatched at the TCL aggregator interface, which is consistent with existing researches [2], [3], [20] and [21]. Reactive power flow and voltage security are handled by the DSO.

Finally, TCLs such as air conditioners, refrigerators and water heaters managed by the same aggregator are assumed to be located at the same bus of the ADN.

B. Optimization Model of Active Distribution Network

1) Uncertainties of Active Distribution Network

The uncertainties in the ADN are mainly incurred by the loads and RESs. Net nodal loads in the ADN under uncertainties can be modeled as:

$$\tilde{P}_{i,t}^{net} = \left(\bar{P}_{i,t}^{L} - \bar{P}_{i,t}^{RES}\right) + \left(\xi_{i,t}^{L,P} - \xi_{i,t}^{RES}\right) = \bar{P}_{i,t}^{net} + \xi_{i,t}^{P} \tag{1}$$

$$\tilde{Q}_{i,t}^{net} = \bar{Q}_{i,t}^{L} + \xi_{i,t}^{L,Q} = \bar{Q}_{i,t}^{net} + \xi_{i,t}^{Q} \tag{2}$$

where $\bar{P}_{i,t}^{L}, \bar{P}_{i,t}^{RES}$ are the expected active power of loads and RESs at i-th node and $\bar{Q}_{i,t}^{L}$ is the expected reactive power of loads at i-th node. $\xi_{i,t}^{L,P}, \xi_{i,t}^{RES}, \xi_{i,t}^{L,Q}$ are the corresponding random prediction errors.

2) Objective of Active Distribution Network

The objective function of DSO aims to minimize the overall operation cost, which includes the generation cost of DGs, the cost of energy purchased from the main grid, the reserve cost from net load uncertainties (namely $f^{DSO}\left(\boldsymbol{x}^{DSO}\right)$), and the operational cost of all TCL aggregators (namely $\sum_{k\in\Omega_{AGG}} f_k^{AGG}\left(\boldsymbol{x}_k^{AGG}\right)$). It is modeled as follows:

$$\min F^{obj} = \min\left[f^{DSO}\left(\boldsymbol{x}^{DSO}\right) + \sum_{k\in\Omega_{AGG}} f_k^{AGG}\left(\boldsymbol{x}_k^{AGG}\right)\right] \tag{3}$$

$$f^{DSO}\left(\boldsymbol{x}^{DSO}\right) = \sum_{t=1}^{T} C^{G}\left(P_{i,t}^{G}\right) + \sum_{t=1}^{T} C^{MG}\left(P_{0,t}^{G}\right) + \sum_{t=1}^{T} C^{R}\left(P_{i,t}^{net}\right) \tag{4}$$

$$C^{G}\left(P_{i,t}^{G}\right) = \sum_{i\in\Omega_G} a_i^{G}\left(P_{i,t}^{G}\right)^2 + b_i^{G} P_{i,t}^{G} + c_i^{G} \tag{5}$$

$$C^{MG}\left(P_{0,t}^{G}\right) = p_t^{P} P_{0,t}^{G} \tag{6}$$

$$C^{R}\left(P_{i,t}^{net}\right) = \sum_{i\in N} p_t^{R}\left(P_{i,t}^{net} - \bar{P}_{i,t}^{net}\right) \tag{7}$$

where $\boldsymbol{x}^{DSO} = \left\{P_{i,t}^{G}, Q_{i,t}^{G}, P_{i,t}^{net}, Q_{i,t}^{net}, \forall i,t\right\}$ are the DSO decision variables. The detailed model of $f_k^{AGG}\left(\boldsymbol{x}_k^{AGG}\right)$ will be illustrated in Section III. Node $\{0\}$ refers to the substation of the AND.

3) Constraints of Active Distribution Network

Firstly, we adopt the convex DistFlow model since it captures constraints of feeder power flows, nodal voltage magnitudes, and power losses as well as providing a feasible solution (global optimum under mild conditions [30]) for the original non-convex OPF. Additionally, considering TCLs often have single-phase connections, the proposed framework is also applicable to the three-phase unbalanced distribution network. The detailed formulation of unbalanced distribution network is further provided in Appendix A.

$$p_{j,t} = \sum_{k:j\to k} P_{jk,t} - \sum_{i:i\to j}\left(P_{ij,t} - r_{ij} l_{ij,t}\right); \forall j\in\mathcal{N}, \forall t\in[T] \tag{8}$$

$$q_{j,t} = \sum_{k:j\to k} Q_{jk,t} - \sum_{i:i\to j}\left(Q_{ij,t} - x_{ij} l_{ij,t}\right) \tag{9}$$

$$v_{j,t} = v_{i,t} - 2\left(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}\right) + \left(r_{ij}^2 + x_{ij}^2\right) l_{ij,t}; \forall (i,j)\in\mathcal{E} \tag{10}$$

$$\left\| \begin{bmatrix} 2P_{ij,t} \\ 2Q_{ij,t} \\ l_{ij,t} - v_{i,t} \end{bmatrix} \right\|_2 \le l_{ij,t} + v_{i,t}; \forall (i,j)\in\mathcal{E}, \forall i\in\mathcal{N} \tag{11}$$

where $P_{ij,t} / Q_{ij,t}$ is the power flow from node i to j. $v_{j,t} = \left|V_{j,t}\right|^2$ and $l_{ij,t} = \left|I_{ij,t}\right|^2$ are defined as the squared voltage and current magnitudes. $z_{ij} = r_{ij} + \boldsymbol{i}x_{ij}$ is the complex impedance on the line $(i,j)\in\mathcal{E}$. Meanwhile, the equality constraints of expected injection power are given by

$$p_{j,t} = P_{j,t}^{G} - P_{j,t}^{net} - P_{j,t}^{AGG} \tag{12}$$

$$q_{j,t} = Q_{j,t}^{G} - Q_{j,t}^{net}. \tag{13}$$

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The nodal voltage and branch current should be constrained within the acceptable values which are modeled by

$$V_{\min}^2 \le v_{i,t} \le V_{\max}^2 \tag{14}$$

$$l_{ij,t} \le I_{\max}^2. \tag{15}$$

Besides, the limits of DGs power generation and power transmission between DN and substation are shown as follows:

$$P_{i,\min}^G \le P_{i,t}^G \le P_{i,\max}^G; \forall i \in \{0\} \cup \Omega_G, \forall t \in [T] \tag{16}$$

$$Q_{i,\min}^G \le Q_{i,t}^G \le Q_{i,\max}^G. \tag{17}$$

The active and reactive power constraints for the net load at each node are modeled as follows:

$$P_{i,t}^{net} \ge \bar{P}_{i,t}^{net}; \forall i \in \mathcal{N}, \forall t \in [T] \tag{18}$$

$$Q_{i,t}^{net} \ge \bar{Q}_{i,t}^{net}; \forall i \in \mathcal{N}, \forall t \in [T]. \tag{19}$$

The operational constraints of *k*-th TCL aggregator can be described by a convex polytope, i.e., a set of linear inequalities:

$$\boldsymbol{P}_j^{AGG} \in \Phi_k^{AGG}, \tag{20}$$

which will be further illustrated in Section III.C. To account for day-ahead prediction errors of net loads, the following JCCs are modeled to ensure the overall robustness of each node in the ADN with a preferable risk level ε_1:

$$\Pr_{\xi^P}\left\{\bar{P}_{i,t}^{net} + \xi_{i,t}^P \le P_{i,t}^{net}, \forall t \in [T]\right\} \ge 1-\varepsilon_1; \forall i \in \mathcal{N} \tag{21}$$

$$\Pr_{\xi^Q}\left\{\bar{Q}_{i,t}^{net} + \xi_{i,t}^Q \le Q_{i,t}^{net}, \forall t \in [T]\right\} \ge 1-\varepsilon_1; \forall i \in \mathcal{N} \tag{22}$$

When the above JCCs are satisfied, the power generation of the system is no less than the practical net loads and any surplus generation is treated as the reserve capacity. Thereby, the JCCs of (21) and (22) ensure the feasibility of (14) and (15).

C. Statistically Feasible Scheduling Model for DSO

For notation simplicity, we rewrite $\xi_{i,t}^{P/Q}$ as $\xi_{i,t}$ uniformly. Besides, we rewrite JCCs of (21) and (22) into a general form:

$$\Pr_{\xi}\left\{C\boldsymbol{\xi} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}\right\} \ge 1-\varepsilon_1 \tag{23}$$

where $\boldsymbol{x}$ is the decision vector. $\boldsymbol{A}$ and $\boldsymbol{C}$ are affine matrices, and $\boldsymbol{b}$ is the constant vector.

The implicit formulation of JCCs in (23) is challenging to solve and the real distribution of $\boldsymbol{\xi}$ is also unknown in practice. In contrast, RO can utilize uncertainty sets to characterize uncertainty, which eliminates the requirement for distribution information and allows us to construct uncertainty sets with *K* historical observations (i.e., $D_\xi = \left\{\hat{\boldsymbol{\xi}}_k\right\}_{k=1}^K$, $\hat{\boldsymbol{\xi}}_k \in R^T$). This motivates us to solve (23) from RO perspective and build the data-driven uncertainty set by historical data. First, consider a conservative approximation of (23) via the RO:

$$C\boldsymbol{\xi} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}, \forall \boldsymbol{\xi} \in U \tag{24}$$

where U is an uncertainty set. By choosing U that covers a $1-\varepsilon_1$ content of $\boldsymbol{\xi}$ (i.e., $\Pr(\boldsymbol{\xi} \in U) \ge 1-\varepsilon_1$), any solution feasible for (24) must satisfy

$$\Pr\left\{C\boldsymbol{\xi} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}\right\} \ge \Pr(\boldsymbol{\xi} \in U) \ge 1-\varepsilon_1, \tag{25}$$

which implies that any feasible solution of (24) using a $1-\varepsilon_1$ content set U is feasible for (23). Furthermore, this formulation provides a general platform to utilize dataset D_ξ to construct the data-driven uncertainty set $U = U(D_\xi)$, which is a $1-\varepsilon_1$ content prediction set for (23) with a prescribed confidence level $(1-\delta)$. Formally, we introduce the definition of statistical feasibility as follows:

Definition 1 (Statistical Feasibility [13]): A solution is statistically feasible if it satisfies the given constraints under the dataset D_ξ with confidence level $1-\delta$. Specifically, for (25), the statistical feasibility is defined as

$$\Pr_{D_\xi}\left\{\Pr\left\{\boldsymbol{\xi} \in U(D_\xi)\right\} \ge 1-\varepsilon_1\right\} \ge 1-\delta. \tag{26}$$

(26) ensures that any solution feasible for (24) is also feasible for (23) with confidence level $1-\delta$ over dataset D_ξ. To make (26) tractable, the following two steps are designed.

First, learn a proper data-driven uncertainty set $U(D_\xi)$ which satisfies $\Pr_{D_\xi}\left\{\Pr\left(\boldsymbol{\xi} \in U(D_\xi)\right) \ge 1-\varepsilon_1\right\} \ge 1-\delta$.

Second, use the learned $U(D_\xi)$ to solve the *robust counterpart* of the original JCCs in (23) by formulating the following inequality:

$$\max_{\xi \in U}\left\{C(j)\boldsymbol{\xi}\right\} \le \boldsymbol{A}(j)\boldsymbol{x} + \boldsymbol{b}(j), \forall j \in [T] \tag{27}$$

1) Statistically Feasible Uncertainty Set Construction

To alleviate the conservativeness of the solution in (26), the set size of constructed $U(D_\xi)$ should be minimized as well as satisfying the statistical feasibility of (26). Thus, a two-phase strategy is adopted. Firstly, we split the dataset D_ξ into two groups, denoted by $D_\xi^a = \left\{\boldsymbol{\xi}_1^a, \ldots, \boldsymbol{\xi}_{n_1}^a\right\}$ and $D_\xi^b = \left\{\boldsymbol{\xi}_1^b, \ldots, \boldsymbol{\xi}_{n_2}^b\right\}$, with sizes n_1 and n_2, respectively. These two data groups are used as follows:

Phase 1: Shape learning. Utilize dataset D_ξ^a to approximate the shape of $U(D_\xi)$. Two common choices of tractable basic geometric shapes are given as follows:

The first is the *ellipsoidal uncertainty set* modeled by:

$$U_1 := \left\{\boldsymbol{\xi} \mid (\boldsymbol{\xi} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{\xi} - \boldsymbol{\mu}) \le s_\rho\right\} \tag{28}$$

where $\boldsymbol{\mu}, \Sigma$ influence the shape and s_ρ characterizes the size of the uncertainty set. $\boldsymbol{\mu}, \Sigma$ can be estimated by dataset D_ξ^a, for instance, setting $\boldsymbol{\mu}$ as the sample mean and Σ as the sample covariance matrix of D_ξ^a. s_ρ is settled in **Phase 2** [13].

The second is the *polytope uncertainty set* with affine matrix $\boldsymbol{D}$ and constant vector $\boldsymbol{d}$:

$$U_2 := \left\{\boldsymbol{\xi} \mid \boldsymbol{D}\boldsymbol{\xi} \le \boldsymbol{d}\right\}. \tag{29}$$

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It should be noted that the construction of ellipsoidal set is intuitive via dataset while that of polytope set is more difficult due to massive parameters of $\boldsymbol{D}$ to be chosen. After determining the shape of $U(D_\xi)$, we need to calculate its size.

Phase 2: Size calibration. In this phase, the D_ξ^b is adopted to calibrate the size of U. The goal is to find the smallest scaling factor s_ρ to satisfy the statistical feasibility of (26).

For the *ellipsoid*, let $g(\boldsymbol{\xi}) = (\boldsymbol{\xi}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{\xi}-\boldsymbol{\mu})$ and (28) is equivalent to $\{\boldsymbol{\xi} \mid g(\boldsymbol{\xi}) \le s_\rho\}$.

For the *polytope*, we firstly add a positive constant vector $\boldsymbol{\tau}$ to ensure that the right-hand side of (29) is larger than zero. Then, U_2 can be rewritten as:

$$U_2 := \left\{ \boldsymbol{\xi} \mid \frac{\boldsymbol{D}(j)\boldsymbol{\xi} + \boldsymbol{\tau}(j)}{\boldsymbol{d}(j) + \boldsymbol{\tau}(j)} \le s_\rho, \forall j \in [T] \right\}. \tag{30}$$

Let

$$g(\boldsymbol{\xi}) = \max_{j \in [T]} \frac{\boldsymbol{D}(j)\boldsymbol{\xi} + \boldsymbol{\tau}(j)}{\boldsymbol{d}(j) + \boldsymbol{\tau}(j)}, \tag{31}$$

and the size calibration of U_2 can also be reformulated as $\{\boldsymbol{\xi} \mid g(\boldsymbol{\xi}) \le s_\rho\}$.

Subsequently, substitute $\boldsymbol{\xi}_i^b \in D_\xi^b$ into $g(\boldsymbol{\xi})$ and all $g(\boldsymbol{\xi}_i^b)$ are sorted in ascending order to obtain $\{g(\boldsymbol{\xi}_i^b)\}_{i=1}^{n_2}$, where $g(\xi_1^b) \le g(\xi_2^b) \le \ldots \le g(\xi_{n_2}^b)$. The desired s_ρ^* is calculated by:

$$\begin{aligned} i^* &= \min\left\{ r : \sum_{k=0}^{r-1} \binom{n_2}{k} (1-\varepsilon)^k \varepsilon^{n_2-k} \ge 1-\delta, 1 \le r \le n_2 \right\} \\ s_\rho^* &= g(\xi_{i^*}^b) \end{aligned}, \tag{32}$$

After constructing statistically feasible uncertainty set, we can reformulate the robust constraints (27) into tractable form by strong duality:

For the *ellipsoid*, the Lagrange function of left-hand side of (27) is:

$$L(\boldsymbol{\xi}, \lambda) = C(j)\boldsymbol{\xi} + \lambda\left(s_\rho^* - (\boldsymbol{\xi}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{\xi}-\boldsymbol{\mu})\right) \tag{33}$$

where λ is the dual variable of constraint (28). Meanwhile, the optimality conditions can be derived as follows:

$$\begin{aligned} (\partial L/\partial \boldsymbol{\xi}) &= C(j)^{\mathrm{T}} - 2\lambda\Sigma^{-1}(\boldsymbol{\xi}-\boldsymbol{\mu}) = 0 \\ (\partial L/\partial \lambda) &= s_\rho^* - (\boldsymbol{\xi}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{\xi}-\boldsymbol{\mu}) = 0 \end{aligned}. \tag{34}$$

Based on (34), we can obtain the analytic expression of $\boldsymbol{\xi}$:

$$\boldsymbol{\xi} = \boldsymbol{\mu} + \sqrt{s_\rho^*} \cdot \Sigma C(j)^{\mathrm{T}} \Big/ \sqrt{C(j)\Sigma C(j)^{\mathrm{T}}}, \tag{35}$$

and the tractable reformulation of (27):

$$C(j)\boldsymbol{\mu} + \sqrt{\left(s_\rho^* C(j)\Sigma C(j)^{\mathrm{T}}\right)} \le \boldsymbol{A}(j)\boldsymbol{x} + \boldsymbol{b}(j). \tag{36}$$

For the *polytope*, the similar derivation can be conducted by strong duality and the tractable reformulation of (27) is:

$$\begin{aligned} &\boldsymbol{\lambda}^T \left(s_\rho^* (\boldsymbol{d}+\boldsymbol{\tau}) - \boldsymbol{\tau}\right) \le \boldsymbol{A}(j)\boldsymbol{x} + b(j) \\ &C(j) = \boldsymbol{\lambda}^T \boldsymbol{D}, \boldsymbol{\lambda} \ge 0 \end{aligned}, \tag{37}$$

where $\boldsymbol{\lambda}$ is the dual vector of constraints (30).

Theorem 1 (Theorem 1 in [13]): Consider an i.i.d. dataset D_ξ, and we partition D_ξ into two sets D_ξ^a and D_ξ^b with sizes n_1 and n_2, respectively. Then, the uncertainty set $U_{1,2}$ built by dataset D_ξ satisfies the statistical feasibility of (26) if $n_2 \ge \log\delta \log^{-1}(1-\varepsilon)$.

Theorem 1 guarantees the statistical feasibility of the constructed uncertainty set in giving a feasible solution under constraints (36) and (37) with confidence $(1-\delta)$ as long as $n_2 \ge \log\delta \log^{-1}(1-\varepsilon)$. However, it remains unclear what is the optimal shape of uncertainty set. Generally, the generic *ellipsoidal uncertainty set* can make the solution overconservative. A better uncertainty set construction can benefit from the specific constraint's form in (23). This idea is illustrated in Fig. 1.

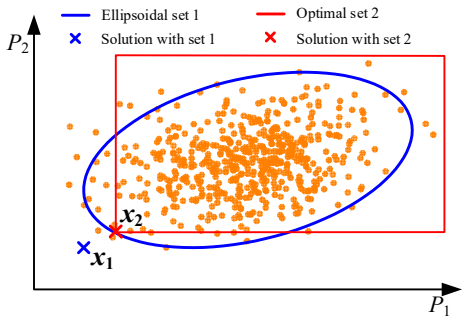

Fig. 1. An illustration of maximum RES utilization under ellipsoidal and optimal uncertainty sets.

In Fig. 1, we maximize RES utilization under uncertainties. The ellipsoidal set (blue curve) generates a conservative solution $\boldsymbol{x}_1$ since $\boldsymbol{x}_1$ needs to be smaller than the worst case at P_1, P_2, respectively. In contrast, due to the separate structure in constraints (21) and (22), the optimal set (red curve) obtains a better solution $\boldsymbol{x}_2$ as well as satisfying the statistical feasibility.

2) Optimal Shape Approximation for Uncertainty Set

Observing (21) and (22), each ξ_j appears separately in $\Pr_\xi\{C\boldsymbol{\xi} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}\}$, such as $\Pr_\xi\{\Lambda\boldsymbol{\xi} \le A\boldsymbol{x} + \boldsymbol{b}\}$ where Λ is a diagonal matrix. Then, the robust counterpart of JCC (27) is equivalent to (38) with the optimal uncertainty set (39):

$$\max_{\xi \in U_{opt}} \boldsymbol{\xi} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}, \tag{38}$$

$$U_{opt} := \{\xi \mid \boldsymbol{\xi} \le \boldsymbol{\xi}^{\max}\}. \tag{39}$$

Next, we need to calculate the base values and scaling factor of $\boldsymbol{\xi}^{\max}$ (i.e., $\boldsymbol{\xi}^{\max} = s_\rho \cdot \boldsymbol{\xi}^{base}$), where $\boldsymbol{\xi}^{base}$ determines the

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shape of U_{opt} and s_ρ controls the size. A simple way to calculate $\boldsymbol{\xi}^{base}$ is using the sample mean (SM):

$$\boldsymbol{\xi}^{base} = \left(\sum\nolimits_{i\in D_\xi^a} \boldsymbol{\xi}_i\right) \Big/ n_1 . \tag{40}$$

The scaling factor s_ρ^* is calibrated by (32) and we obtain $\boldsymbol{\xi}^{SM} = s_\rho^* \cdot \boldsymbol{\xi}^{base}$. However, the performance of $\boldsymbol{\xi}^{SM}$ would depend on the circumstances since it does not consider the specific optimization problem of the DSO. Thereby, we propose an OSA method to estimate $\boldsymbol{\xi}^{\max}$ as follows:

Phase 1: Optimal Shape learning.

Firstly, define the decision variable $\boldsymbol{\xi}^{opt} \in R^T$ to determine the optimal shape of $\boldsymbol{\xi}^{base}$, which is a function in terms of sample set D_ξ, denoted by $f_\xi\left(D_\xi\right)$. T boundary points of $\boldsymbol{\xi}^{opt}$ are defined by $\left\{\boldsymbol{\xi}_j^{bound}, \forall j \in [T]\right\}$ to constrain the minimum value at j-th component of $\boldsymbol{\xi}^{opt}$.

Secondly, calculate $\boldsymbol{\xi}_j^{bound}$. The j-th component of $\boldsymbol{\xi}_j^{bound}$ (denoted by $\boldsymbol{\xi}_{j,j}^{bound}$) is determined by calculating the desired boundary value for j-th ICC under the given risk level ε_1. The other $\boldsymbol{\xi}_{j,i}^{bound}, \forall i \neq j$ are set as the largest values among samples. Specifically, for $\boldsymbol{\xi}_{j,j}^{bound}$, we sort $D_\xi^a = \left\{\boldsymbol{\xi}_1^a, \ldots, \boldsymbol{\xi}_{\boldsymbol{n}_1}^a\right\}$ in terms of j-th component in ascending order to obtain $\left\{\xi_{j,k}^{sort}\right\}_{k=1}^{n_1}$, where $\xi_{j,1}^{sort} \le \xi_{j,2}^{sort} \le \ldots \le \xi_{j,n_1}^{sort}$. The desired $\boldsymbol{\xi}_{j,j}^{bound}$ is calculated by:

$$\begin{aligned} j^* &= \min\left\{ r : \sum_{k=0}^{r-1} \binom{n_1}{k} (1-\varepsilon)^k \, \varepsilon^{n_1-k} \ge 1-\delta \right\} \\ \boldsymbol{\xi}_{j,j}^{bound} &= \xi_{j,j^*}^{sort} \end{aligned} \tag{41}$$

For $\boldsymbol{\xi}_{j,i}^{bound}, \forall i \neq j$, the desired $\boldsymbol{\xi}_{j,i}^{bound}$ is equal to ξ_{i,n_1}^{sort} by sorting $D_\xi^a = \left\{\boldsymbol{\xi}_1^a, \ldots, \boldsymbol{\xi}_{\boldsymbol{n}_1}^a\right\}$ in terms of i-th component.

Thirdly, after obtaining boundary points $\left\{\boldsymbol{\xi}_j^{bound}, \forall j \in [T]\right\}$, they are combined with the point $\boldsymbol{\xi}^{SM}$ to determine T hyperplanes for the linear convex approximation for $\boldsymbol{\xi}^{opt}$ and the feasible region of $\boldsymbol{\xi}^{opt}$ can be described as:

$$\boldsymbol{n}_l^T \boldsymbol{\xi}^{opt} + \zeta_l \ge 0, \forall l \in [T] \tag{42}$$

where $\boldsymbol{n}_l \in R^T, \zeta_l$ are the normal vector and offset of l-th hyperplanes. Based on the above three steps, we can formulate the OSA optimization to learn the optimal $\boldsymbol{\xi}^{opt}$.

Proposition 1: the OSA-based shape optimization for $\boldsymbol{\xi}^{opt}$ is equivalent to the following tractable convex problem:

$$\begin{aligned} &\min_{\boldsymbol{x}^{DSO}, \boldsymbol{\xi}^{opt}} \text{ objective function (3)} \\ &\text{s.t. constraints (8)-(20)} \\ &\boldsymbol{\xi}^{opt} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \\ &\boldsymbol{n}_l^T \boldsymbol{\xi}^{opt} + \zeta_l \ge 0, \forall l \in [T] \\ &\boldsymbol{\xi}_j^{opt} \ge \boldsymbol{\xi}_{j,j}^{bound}, \forall j \in [T] \end{aligned} \tag{43}$$

Proof: First, the original JCCs of DSO optimization are reformulated by robust counterpart $\boldsymbol{\xi}^{opt} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$ with decision vector $\boldsymbol{\xi}^{opt}$ to be optimized. Secondly, (42) constrains the feasible origin via the intersection of T half-spaces. Finally, constraint $\boldsymbol{\xi}_j^{opt} \ge \boldsymbol{\xi}_{j,j}^{bound}$ indicates that the solution of $\boldsymbol{\xi}^{opt}$ also satisfies j-th ICC under the given risk level ε_1 with statistical feasibility. Through (43), we can approximate the optimal $\boldsymbol{\xi}^{opt}$.

Remark 1: In (43), we add T cutting planes to approximate the function of $\boldsymbol{\xi}^{opt}$ in terms of sample set D_ξ. To improve the accuracy, more sampled points can be considered to form more cutting planes. In general, the proposed OSA method is a general way to approximate $\boldsymbol{\xi}^{opt}$. After obtaining $\boldsymbol{\xi}^{opt}$, we need to calibrate its size to ensure it satisfies the statistical feasibility, which is elaborated in **Phase 2**.

Phase 2: Size calibration. In this phase, the D_ξ^b is adopted to calibrate the size of U. The goal is to find the smallest scaling factor s_ρ to satisfy the statistical feasibility of (26). To achieve this, first let $\boldsymbol{\xi}^{base} = \boldsymbol{\xi}^{opt}$ for notation uniform and rewrite the optimal uncertainty set (39) as:

$$U_{opt} := \left\{ \boldsymbol{\xi} \mid \frac{\boldsymbol{\xi}(j) + \boldsymbol{\tau}(j)}{\boldsymbol{\xi}^{base}(j) + \boldsymbol{\tau}(j)} \le s_\rho, \forall j \in [T] \right\}, \tag{44}$$

which is similar to the procedure of (30). Then, let

$$g(\boldsymbol{\xi}) = \max_{j\in[T]} \frac{\boldsymbol{\xi}(j) + \boldsymbol{\tau}(j)}{\boldsymbol{\xi}^{base}(j) + \boldsymbol{\tau}(j)}, \tag{45}$$

and (44) is equivalent to $\left\{\boldsymbol{\xi} \mid g(\boldsymbol{\xi}) \le s_\rho\right\}$. Substitute $\boldsymbol{\xi}_i^b \in D_\xi^b$ into $g(\boldsymbol{\xi})$ and all $g\left(\boldsymbol{\xi}_i^b\right)$ are sorted in ascending order. The desired s_ρ^* is calculated by Eq.(32). Finally, we obtain $\boldsymbol{\xi}^{\max} = s_\rho^* \cdot \boldsymbol{\xi}^{base}$, which also satisfies the statistical feasibility.

3) Tractable Formulation under Constructed Uncertainty Set

After determining the optimal uncertainty set U_{opt}, the robust constraint (27) should be reformulated in a tractable form with the following proposition.

Proposition 2: With the optimal uncertainty set of (44), the robust constraint (38) can be transformed into the following linear constraints:

$$s_\rho^*\left(\boldsymbol{\xi}^{base} + \boldsymbol{\tau}\right) - \boldsymbol{\tau} \le \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \tag{46}$$

Proof: the robust constraint (38) with the optimal uncertainty set is equivalent to:

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$$\begin{aligned}&\max_{\xi\in U_{opt}} \boldsymbol{\xi} \le \boldsymbol{Ax}+\boldsymbol{b}\\&U_{opt}=\left\{\boldsymbol{\xi}\mid \boldsymbol{\xi}+\boldsymbol{\tau}\le s_\rho^*\left(\boldsymbol{\xi}^{base}+\boldsymbol{\tau}\right)\right\}\end{aligned}. \tag{47}$$

Let $\boldsymbol{\xi}=s_\rho^*\left(\boldsymbol{\xi}^{base}+\boldsymbol{\tau}\right)-\boldsymbol{\tau}$ to maximize $\boldsymbol{\xi}$ and we obtain (46), which completes the proof.

Finally, the tractable DSO model is summarized as follows:

$$\begin{aligned}&\min \text{ objective function (3)}\\&\text{s.t. constraints (8)-(20)}\\&\quad\text{linear constraints (46)}\end{aligned} \tag{48}$$

III. Mathematic Model of TCL Optimization with Model Predictive Control based Aggregation

In this section, we first derive the feasible region model of individual TCLs. Then, we employ the polytope-based aggregation with statistically feasible uncertainty set to build the TCL aggregator under ambient uncertainties. Finally, a MPC module is embedded in the TCL aggregator to improve feasible capacities.

A. Dynamic Model of TCLs

In this paper, we focus on air conditioner (AC) loads in the cooling mode due to their widespread use and significant impact on peak load response. Nevertheless, the proposed model can be extended to other types of TCLs (e.g., heating and water heaters). The linear model is utilized to approximate the temperature dynamics of TCLs in the cooling mode, which is commonly used in the literature [31].

$$\theta_{i,t+1}=\left(1-\frac{\Delta t}{R_iC_i}\right)\theta_{i,t}+\left(\frac{1}{R_iC_i}\theta_t^{out}-\kappa_i^{trans}\frac{\eta_i}{C_i}p_{i,t}^{TCL}\right)\Delta t \tag{49}$$

where R_i, C_i are the thermal capacitance and resistance of *i-th* TCL. $p_{i,t}^{TCL}\in\left[0,p_{i,\max}^{TCL}\right]$ is the active power consumption with its coefficient of performance η_i. $\kappa_i^{trans}\in\{-1,1\}$ is an indicator of TCL operation mode, which equals 1 for cooling TCLs and -1 for heating TCLs. In the following, we omit κ_i^{trans} since we focus on cooling TCLs. To satisfy users' comfort, the indoor temperature of TCL is constrained by $\left[\theta_{i,r}-\Delta\theta_i,\theta_{i,r}+\Delta\theta_i\right]$, where $\theta_{i,r},\Delta\theta_i$ are user specified temperature set-point and half of the temperature dead-band. Furthermore, we introduce the virtual battery (VB) model to represent the flexibility of TCLs in an intuitive way:

$$\Phi_i=\left\{\boldsymbol{u}_i\in R^T\left|\begin{array}{l}x_{i,t+1}=a_ix_{i,t}+u_{i,t}\Delta t,\forall t\in[T]\\0\le x_{i,t}\le\dfrac{2C_i\Delta\theta_i}{\eta_i}=x_{i,t}^+,\forall t\in[T]\\u_{i,t}^-=p_{i,t}^{out}-p_{i,\max}^{TCL}\le u_{i,t}\le p_{i,t}^{out}=u_{i,t}^+\end{array}\right.\right\} \tag{50}$$

with decision variable $u_{i,t}=p_{i,t}^{out}-p_{i,t}^{TCL}$ and state variable $x_{i,t}=C_i\left(\theta_{i,t}-\theta_{i,\min}\right)/\eta_i$. The equivalent outdoor power disturbance is $p_{i,t}^{out}=\left(\theta_t^{out}-\theta_{i.\min}\right)/b_i$. $a_i=1-\Delta t/\left(R_iC_i\right)$ and $b_i=\eta_iR_i$. Further, the feasible region of (50) can be rewritten in the form of the H-representation of a polytope:

$$\Phi_i=\left\{\boldsymbol{u}_i\in R^T\left|\boldsymbol{H}_i\boldsymbol{u}_i\le\boldsymbol{h}_i\right.\right\} \tag{51}$$

where $\boldsymbol{H}_i=\left[I_T,\Lambda_i^{-1}\Delta t,-I_T,-\Lambda_i^{-1}\Delta t\right]^T$. I_T is identity matrix. $\boldsymbol{h}_i=[\boldsymbol{u}_i^+,\boldsymbol{x}_i^+-\Lambda_i^{-1}\boldsymbol{c}_i,-\boldsymbol{u}_i^-,\Lambda_i^{-1}\boldsymbol{c}_i]^T$. $\Lambda_i=I_T+\mathrm{diag}\left(-a_i;-1\right)$ with $\boldsymbol{c}_i=\left[a_iC_i\left(\theta_{i,0}-\theta_{\min}\right)/\eta_i,0,\ldots,0\right]^T$.

B. Polytope-Based Aggregation under Ambient Uncertainties

In this paper, we focus on day-ahead TCL aggregation, referring to the controllable TCLs that have explicitly enrolled in direct load control (DLC) programs and accepted the associated control and compensation terms. Thereby, we only consider the uncertainty of outdoor temperature. For the uncertain social behaviors of consumers (e.g., "opt-out" options [32] and DR statuses [33]), they are widely modeled in the real-time operation and not considered in this paper.

The M-sum of polytopes $\Phi_k^{agg}=\biguplus_i\Phi_{k,i}$ is a direct way to calculate the aggregated flexible capacities of TCLs, where *i*, *k* are indices for *i*-th TCL in *k*-th aggregator. However, the M-sum is computational intractable since it is a NP-hard problem. Thereby, the polytope-based method is introduced to efficiently approximate the M-sum [20].

Given a prototype set $\Phi_k^{base}=\left\{\boldsymbol{u}_k^{base}\left|\boldsymbol{H}_k^{base}\boldsymbol{u}_k^{base}\le\boldsymbol{h}_k^{base}\right.\right\}$, where its parameters $\boldsymbol{H}_k^{base},\boldsymbol{h}_k^{base}$ can be chosen as the average values of all $\boldsymbol{H}_{k,i},\boldsymbol{h}_{k,i}$ in *k*-th aggregator. Then, the flexible region of $\Phi_{k,i}$ can be approximated by a homothet of Φ_k^{base}. Unlike traditional methods using scalar scaling factor $\beta_{k,i}$ (i.e., the scaling in all dimensions is uniform in [20]), this paper adopts affine transformation in the form of $\hat{\Phi}_{k,i}=\boldsymbol{B}_{k,i}\Phi_k^{base}+\boldsymbol{t}_{k,i}$ to approximate the original polytope, which is first developed in [22]. $\boldsymbol{B}_{k,i}\in R^{T\times T},\boldsymbol{t}_{k,i}\in R^T$ are affine scaling matrix and translation vector, respectively. Moreover, the M-sum of approximated homothets can be calculated efficiently by:

$$\hat{\Phi}_k^{agg}=\biguplus_i\hat{\Phi}_{k,i}=\left(\sum\nolimits_i\boldsymbol{B}_{k,i}\right)\Phi_k^{base}+\sum\nolimits_i\boldsymbol{t}_{k,i}\subseteq\Phi_k^{agg} \tag{52}$$

To achieve the maximization of inner-approximation area considering ambient uncertainties integrated into the random variable θ_t^{out}, the following JCC-based optimization with risk level ε_2 is formulated to determine the values of $\boldsymbol{B}_{k,i},\boldsymbol{t}_{k,i}$:

$$\begin{aligned}&\max_{\boldsymbol{B}_{k,i},\boldsymbol{t}_{k,i}}\left|\det\left(\boldsymbol{B}_{k,i}\right)\right|\\&s.t.\Pr_{\boldsymbol{\theta}^{out}}\left\{\boldsymbol{B}_{k,i}\Phi_k^{base}+\boldsymbol{t}_{k,i}\subseteq\Phi_{k,i}\right\}\ge1-\varepsilon_2;\boldsymbol{B}_{k,i}\succ0\end{aligned} \tag{53}$$

To make (53) tractable, the objective function is firstly linearized by the first-order Taylor expansion (i.e., the trace of $\boldsymbol{B}_{k,i}$, $\mathrm{tr}\left(\boldsymbol{B}_{k,i}\right)$). Secondly, let $\boldsymbol{u}_{k,i}=\boldsymbol{B}_{k,i}\breve{\boldsymbol{u}}_{k,i}^{base}+\boldsymbol{t}_{k,i}$ where $\breve{\boldsymbol{u}}_{k,i}^{base}$ is a middle variable. We have:

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$$\breve{\Phi}_{k,i}^{base} = \left\{ \breve{\boldsymbol{u}}_{k,i}^{base} \in R^T \middle| \boldsymbol{H}_i \boldsymbol{B}_{k,i} \breve{\boldsymbol{u}}_{k,i}^{base} \le \boldsymbol{h}_i - \boldsymbol{H}_i \boldsymbol{t}_{k,i} \right\}$$
$$\Phi_{k,i} = \boldsymbol{B}_{k,i} \breve{\Phi}_{k,i}^{base} + \boldsymbol{t}_{k,i} \tag{54}$$

Combining Eq.(54) with $\boldsymbol{B}_{k,i}\Phi_k^{base} + \boldsymbol{t}_{k,i} \subseteq \Phi_{k,i}$, we have $\Phi_k^{base} \subseteq \breve{\Phi}_{k,i}^{base}$. Based on Farkas' Lemma [20], the problem (53) can be further reformulated as follows:

$$\begin{aligned} &\max_{\boldsymbol{B}_{k,i},\boldsymbol{t}_{k,i},\boldsymbol{G}_{k,i}} \left| \mathrm{tr}\left(\boldsymbol{B}_{k,i}\right) \right| \\ &s.t. \boldsymbol{G}_{k,i}\boldsymbol{H}_k^{base} = \boldsymbol{H}_{k,i}\boldsymbol{B}_{k,i}; \\ &\quad \Pr_{\boldsymbol{\theta}^{out}} \left\{ -\boldsymbol{h}_{k,i} \le -\boldsymbol{G}_{k,i}\boldsymbol{h}_k^{base} - \boldsymbol{H}_{k,i}\boldsymbol{t}_{k,i} \right\} \ge 1-\varepsilon_2; \\ &\quad \boldsymbol{B}_{k,i} \succ 0; \boldsymbol{G}_{k,i} \succ 0. \end{aligned} \tag{55}$$

where the random vector $\boldsymbol{\theta}^{out}$ only exist in $\boldsymbol{h}_{k,i}$ and appears separately in each constraint.

Remark 2: the JCC-based flexible region maximization of (55) can be transformed into the linear program by **Proposition 2** with the optimal uncertainty set. However, the direct *T*-step aggregation of TCLs optimized by (55) might lose considerable flexible capacities since the practical flexible range in (50) at time *t* is the function of control bounds (e.g., $u_{i,t}^{+/-}$) and current states (e.g., $x_{i,t}$). In the following subsection, we propose a MPC-based aggregation method to further improve the flexible capacities of TCL aggregators.

C. MPC-based Rolling Optimization of TCL Aggregator

In the day-ahead scheduling, the objective of TCL aggregator is to minimize the energy cost minus the revenue of providing flexible reserves according to the prices offered by DSO [34]:

$$\min f_k^{AGG}\left(\boldsymbol{x}_k^{AGG}\right) = \sum_{t=1}^{T} C_k^{AGG}\left(x_{k,t}^{AGG}\right) \tag{56}$$

$$C_k^{AGG}\left(x_{k,t}^{AGG}\right) = \pi_t^P P_{k,t}^{AGG} - \pi_t^{Ru} Ru_{k,t}^{AGG} - \pi_t^{Rd} Rd_{k,t}^{AGG} \tag{57}$$

where $x_{k,t}^{AGG} = \left\{ P_{k,t}^{AGG}, Ru_{k,t}^{AGG}, Rd_{k,t}^{AGG}, u_{k,t}^{base}, up_{k,t}^{base}, ud_{k,t}^{base} \right\}$ are decision variables of *k*-th aggregator for $\forall t \in [T]$. $\pi_t^{P,Ru,Rd}$ are the energy and reserve offering prices for aggregators [34].

As mentioned in *Remark 2*, to improve the flexible range of TCL aggregator, the states of TCLs should be updated at each time step *t*. Thus, the MPC-based rolling optimization for TCL aggregator *k* is further developed as follows:

$$\min f_k^{AGG}\left(\boldsymbol{x}_k^{AGG}\right) = \sum_{\tau=t}^{t+m} \left[C_k^{AGG}\left(x_{k,\tau}^{AGG}\right) \right], \forall t \in [T] \tag{58}$$

where $m = \min\left(T_{mpc}, T-t\right)$ is the MPC horizon. Meanwhile, $\boldsymbol{H}_k^{base}$ and $\boldsymbol{h}_k^{base}$ are updated dynamically based on optimization results before time step *t* and their dimensions become m. Then, the operation constraints of TCL aggregator are given by:

$$\boldsymbol{H}_k^{base}\left(1:2m,:\right)\boldsymbol{u}_k^{base} + \boldsymbol{up}_k^{base} \le \boldsymbol{h}_k^{base}\left(1:2m,:\right) \tag{59}$$

$$\boldsymbol{H}_k^{base}\left(2m+1:4m,:\right)\boldsymbol{u}_k^{base} + \boldsymbol{ud}_k^{base} \le \boldsymbol{h}_k^{base}\left(2m+1:4m,:\right) \tag{60}$$

$$\boldsymbol{P}_k^{AGG} = \sum\nolimits_i \boldsymbol{p}_{k,i}^{out} - \left(\sum\nolimits_i \boldsymbol{B}_{k,i}\right)\boldsymbol{u}_k^{base} - \sum\nolimits_i \boldsymbol{t}_{k,i} \tag{61}$$

$$\boldsymbol{Ru}_k^{AGG} = \left(\sum\nolimits_i \boldsymbol{B}_{k,i}\right)\boldsymbol{up}_k^{base}; \boldsymbol{Rd}_k^{AGG} = \left(\sum\nolimits_i \boldsymbol{B}_{k,i}\right)\boldsymbol{ud}_k^{base} \tag{62}$$

where bold letters denote the compact form with *m*-dimension. By solving problem (58), decisions $x_{k,\tau}^{AGG}, \forall \tau \in [t, t+m]$ can be obtained. Finally, by calculating (63), the TCL aggregator can directly allocate the dispatching signals to individual TCLs.

$$\boldsymbol{p}_{k,i}^{TCL} = \boldsymbol{p}_{k,i}^{out} - \left(\boldsymbol{B}_{k,i}\boldsymbol{u}_k^{base} + \boldsymbol{t}_{k,i}\right) \tag{63}$$

Since the time horizon of TCL aggregation becomes *m* under MPC framework, we change the JCC formulation of (55) into ICC considering the more accurate prediction of $\boldsymbol{\theta}^{out}$ than net loads. Furthermore, due to the separate structure of CCs in terms of $\boldsymbol{\theta}^{out}$, we can determine the optimal uncertainty set and tractable CC reformulation in (55) conveniently. Meanwhile, we can also calibrate the value of each $\boldsymbol{\xi}_j^{\max}$ based on (32). In summary, the MPC-based rolling optimization of TCLs is illustrated in **Algorithm 1**.

Algorithm 1: MPC-based rolling optimization of TCLs

Input: *t*=1. Initial states of individual TCLs $\left\{x_{k,i,0}; \forall i,k\right\}$. Determine the value of $\boldsymbol{\xi}^{\max}$ by $\boldsymbol{\xi}^{\max} = s_\rho^* \cdot \boldsymbol{\xi}^{opt}$.

for $t = 1:T$, **do**

S1: MPC horizon is $m = \min\left(T_{mpc}, T-t\right)$.

S2: Update parameters $\boldsymbol{H}_k^{base}, \boldsymbol{h}_k^{base}$ of the prototype for *k*-th aggregator in *m* dimensions.

S3: Solve TCL inner approximation problem in (55) with tractable reformulation by (38). Obtain $\boldsymbol{B}_{k,i}, \boldsymbol{t}_{k,i}$ of *i*-th TCL in *k*-th aggregator in *m* dimensions.

S4: Solve TCL aggregator scheduling problem in (58) with constraints (59)-(62). Obtain $x_{k,\tau}^{AGG}; \forall \tau \in [t, t+m]$.

S5: Disaggregate dispatching signals to each TCL by (63) to obtain $p_{k,i,\tau}^{TCL}; \forall \tau \in [t, t+m]$.

S6: Update the state of TCL: $x_{k,i,t+1} = a_{k,i}x_{k,i,t} + u_{k,i,t}\Delta t$

S7: Only save the first step result $x_{k,\tau}^{AGG}|_{\tau=t}$.

Output: the day-ahead dispatching results: $x_{k,t}^{AGG}, \forall t \in [T]$.

IV. Asynchronous Distribute Scheduling Framework

In this section, we divide the centralized scheduling problem of ADN into subproblems of DSO and TCL aggregators. Only aggregated power and reserve signals are interacted between DSO and aggregators, which relieves the communication burden of DSO as well as preserving privacy of individual TCLs. The asynchronous R-ADMM is further developed to solve the reformulated problem in a distributed way. The overall framework is illustrated in Fig.2.

A. Synthesis of Centralized Problem

The whole scheduling problem of ADN can be synthesized and rewritten into the following centralized problem:

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This article has been accepted for publication in IEEE Transactions on Power Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPWRS.2026.3670270

$$\min_{\boldsymbol{x}^{DSO},\boldsymbol{x}_k^{AGG}}\left[f^{DSO}\left(\boldsymbol{x}^{DSO}\right)+\sum_{k\in\Omega_{AGG}}f_k^{AGG}\left(\boldsymbol{x}_k^{AGG}\right)\right]$$
$$\text{s.t. DSO constraints (8)-(19),(46)} \tag{64}$$
$$\text{AGG constraints (59)-(62)}$$

where $\boldsymbol{P}_k^{AGG},\forall k\in\Omega_{AGG}$ is the coupled decision vector between DSO and TCL aggregators. Nevertheless, solving the above centralized problem requires that TCL aggregators submit adequate information to the DSO, which brings communication challenges to DSO and jeopardizes the privacy of TCL users. Therefore, the distributed method needs to be elaborated to address these challenges.

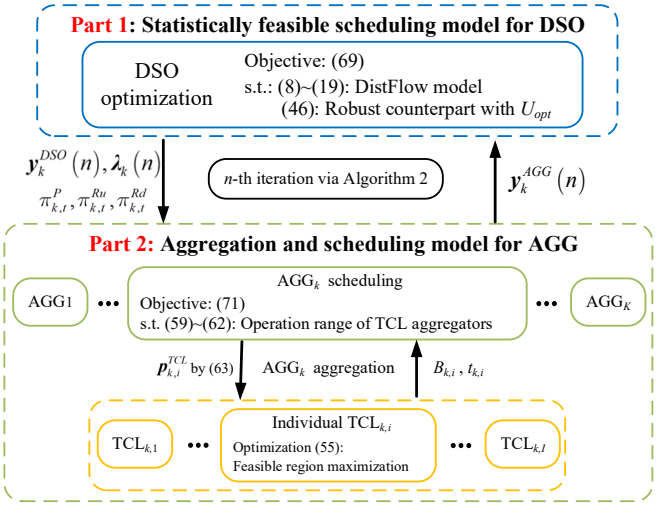

Fig. 2. The distributed framework for the scheduling optimization of ADN with TCL aggregators.

B. R-ADMM based Distributed Method

In the proposed distributed scheduling framework of Fig. 2, the DSO is responsible for solving optimal ADN power flow, and TCL aggregators calculate their flexible capacities and determine their bidding decisions under uncertainties. The consensus variables $\boldsymbol{y}_k=\left\{P_{k,t}^{AGG},\forall t\in[T]\right\}$ are co-optimized by the DSO and TCL aggregators.

To solve the distributed coordination problem between the DSO and TCL aggregators, the ADMM-based methods are widely adopted to solve such a problem. However, it confronts with the risks of lossy communications and slow convergence rate with increasing participants. To tackle this, we adopt the R-ADMM method, which is a generalized version of ADMM based on *Peaceman-Rachford* (PR) splitting with a relaxed factor $\alpha\in(0,1)$. To apply R-ADMM method, we define the following two convex functions:

$$F^{DSO}\left(\boldsymbol{y}_k^{DSO}\right):=\min_{\boldsymbol{x}^{DSO}}f^{DSO}\left(\boldsymbol{x}^{DSO}\right)\ \text{s.t.}\ \boldsymbol{g}^{DSO}\left(\boldsymbol{x}^{DSO},\boldsymbol{y}_k^{DSO}\right)\le 0 \tag{65}$$

$$\begin{aligned}F_k^{AGG}\left(\boldsymbol{y}_k^{AGG}\right)&:=\min_{\boldsymbol{x}_k^{AGG}}f_k^{AGG}\left(\boldsymbol{x}_k^{AGG},\boldsymbol{y}_k^{AGG}\right)\\ &\text{s.t.}\ \boldsymbol{g}_k^{AGG}\left(\boldsymbol{x}_k^{AGG},\boldsymbol{y}_k^{AGG}\right)\le 0\end{aligned} \tag{66}$$

where $\boldsymbol{g}^{DSO},\boldsymbol{g}_k^{AGG}$ are operation constraints of DSO and k-th TCL aggregator. $\boldsymbol{y}_k^{DSO},\boldsymbol{y}_k^{AGG}$ are the estimates for $\boldsymbol{y}_k$ by the DSO and k-th aggregator. Then, (64) can be rewritten as:

$$\begin{aligned}&\min_{\boldsymbol{y}_k^{DSO},\boldsymbol{y}_k^{AGG}}F^{DSO}\left(\boldsymbol{y}_k^{DSO}\right)+\sum_{k\in\Omega_{AGG}}F_k^{AGG}\left(\boldsymbol{y}_k^{AGG}\right)\\ &\text{s.t.}\ \boldsymbol{y}_k^{DSO}-\boldsymbol{y}_k^{AGG}=0;\forall k\in\Omega_{AGG}\end{aligned} \tag{67}$$

Define the Lagrangian of (67) as

$$\begin{aligned}L_\rho\left(\boldsymbol{y}_k^{DSO},\boldsymbol{y}_k^{AGG},\boldsymbol{\lambda}_k\right):=&F^{DSO}\left(\boldsymbol{y}_k^{DSO}\right)+\sum_{k\in\Omega_{AGG}}F_k^{AGG}\left(\boldsymbol{y}_k^{AGG}\right)\\ &+\sum_{k\in\Omega_{AGG}}\left[\frac{\rho}{2}\left\|\boldsymbol{y}_k^{DSO}-\boldsymbol{y}_k^{AGG}\right\|^2-\left\langle\boldsymbol{\lambda}_k,\boldsymbol{y}_k^{DSO}-\boldsymbol{y}_k^{AGG}\right\rangle\right]\end{aligned} \tag{68}$$

where $\rho>0$ is the penalty factor and $\boldsymbol{\lambda}_k$ is the dual multipliers in (67). Then, by applying the relaxed PR splitting to (68), the R-ADMM algorithm can be derived as:

$$\boldsymbol{y}_k^{DSO}(n+1)=\arg\min_{\boldsymbol{y}_k^{DSO}}\begin{bmatrix}L_\rho\left(\boldsymbol{y}_k^{DSO},\boldsymbol{y}_k^{AGG}(n),\boldsymbol{\lambda}_k(n)\right)\\ +\rho(2\alpha-1)\left\langle\boldsymbol{y}_k^{DSO},\boldsymbol{y}_k^{DSO}(n)-\boldsymbol{y}_k^{AGG}(n)\right\rangle\end{bmatrix} \tag{69}$$

$$\boldsymbol{\lambda}_k(n+1)=\boldsymbol{\lambda}_k(n)-\rho\begin{bmatrix}\left(\boldsymbol{y}_k^{DSO}(n+1)-\boldsymbol{y}_k^{AGG}(n)\right)\\ +(2\alpha-1)\left(\boldsymbol{y}_k^{DSO}(n)-\boldsymbol{y}_k^{AGG}(n)\right)\end{bmatrix} \tag{70}$$

$$\boldsymbol{y}_k^{AGG}(n+1)=\arg\min_{\boldsymbol{y}_k^{AGG}}\left[L_\rho\left(\boldsymbol{y}_k^{DSO}(n+1),\boldsymbol{y}_k^{AGG},\boldsymbol{\lambda}_k(n+1)\right)\right] \tag{71}$$

Define the *primal residual* and *dual residual* as:

$$\begin{aligned}\boldsymbol{r}_k(n)&=\left(\boldsymbol{y}_k^{DSO}(n)-\boldsymbol{y}_k^{AGG}(n)\right)/\boldsymbol{y}_k^{DSO}(n)\\ \boldsymbol{s}_k(n)&=\left(\boldsymbol{y}_k^{DSO}(n)-\boldsymbol{y}_k^{DSO}(n-1)\right)/\boldsymbol{y}_k^{DSO}(n)\end{aligned}. \tag{72}$$

The stopping criteria are: $\max\left(\left\|\boldsymbol{r}_k(n)\right\|_2,\left\|\boldsymbol{s}_k(n)\right\|_2\right)>\varepsilon^{err}$. Meanwhile, to speed up the iteration, an increasing factor $\gamma>1$ is used to update $\rho_{n+1}=\gamma\rho_n$ when $\left\|\boldsymbol{r}_k(n)\right\|_2>\left\|\boldsymbol{s}_k(n)\right\|_2$.

The synchronous updating version of (69) confronts slow convergence speed when the interaction between the DSO and TCL aggregators encounters communication delays and asynchrony since different agents have different computational time and local clocks. Thus, an asynchronous variant is proposed in this paper. Firstly, we describe the communication asynchrony following [35]. The local clock C_k is the set of time instants when k-th participant finishes solving its subproblem with the received consensus variables from others in the last time. If $n\notin C_k$, it keeps the solution unchanged (i.e., $x_k(n+1)=x_k(n)$). Then, the global clock is defined as $C_g=C_0\cup C_1\cup\ldots\cup C_{|\Omega_{agg}|}$ and C_0 is the DSO's clock. Meanwhile, we further define $\delta_{0k}^n,\delta_{k0}^n\in\left[0,n_d^{\max}\right]$ as the time delay associated with the DSO and k-th aggregator at $n\in C_g$. $n_d^{\max}$ is the maximum time delay.

To overcome communication asynchrony, when $n\notin C_k$, DSO and AGG$_k$ keep their solutions unchanged. Otherwise,

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they solve their subproblems based on the information received from others in the last time. Finally, the asynchronous R-ADMM method is presented in **Algorithm 2**.

Algorithm 2: Asynchronous R-ADMM algorithm

Input: Set the iteration step n=0, error tolerance ε^{err}, penalty factor ρ_0, increasing factor γ, relaxed factor α. Initialize consensus variables $\boldsymbol{y}_k^{DSO}(0), \boldsymbol{y}_k^{AGG}(0)$.

While $\max\left(\left\|\boldsymbol{r}_k(n)\right\|_2, \left\|\boldsymbol{s}_k(n)\right\|_2\right) > \varepsilon^{err}$ **do**

S1: $n = n+1$.

S2: If $n \in C_0$, call (69) and (70) to update and broadcast $\boldsymbol{y}_k^{DSO}(n), \boldsymbol{\lambda}_k(n)$ to k-th TCL aggregator. Otherwise, $\boldsymbol{y}_k^{DSO}(n) = \boldsymbol{y}_k^{DSO}(n-1)$, $\boldsymbol{\lambda}_k(n) = \boldsymbol{\lambda}_k(n-1)$.

S3: If $n \in C_k$, call (71) to update and send $\boldsymbol{y}_k^{AGG}(n)$ to the DSO. Otherwise, $\boldsymbol{y}_k^{AGG}(n) = \boldsymbol{y}_k^{AGG}(n-1)$.

S4: Update residuals $\left\|\boldsymbol{r}_k(n)\right\|_2$, $\left\|\boldsymbol{s}_k(n)\right\|_2$.

S5: If $n \in C_0$ and $\left\|\boldsymbol{r}_k(n)\right\|_2 > \left\|\boldsymbol{s}_k(n)\right\|_2$, update $\rho_n = \gamma\rho_{n-1}$. Otherwise, $\rho_n = \rho_{n-1}$.

End While

V. Case Studies

In this section, the proposed model and algorithm are evaluated through case studies on the IEEE 33-bus and the 123-bus distribution networks. The codes are implemented in Python environment and solved by Gurobi 11.0.2 on a PC with an Intel Core i7-12700H CPU@ 2.30GHz and 16GB RAM.

A. IEEE 33-Bus System Test

The modified IEEE 33-bus system is with 2 controllable DGs, 3 TCL aggregators and 2 RES units, as shown in Fig. 3. The energy and reserve prices derived from the electricity market of China Southern Power Grid Co., Ltd. are given in [36]. To determine the data-driven uncertainty sets, we use Monte Carlo simulation to generate 1200 samples of loads and RESs from mixed normal and uniform distributions. The standard deviation of load and RES uncertainties is set 10% of forecasted power and the risk level ε_1 is 5%. The confidence level $1-\delta$ is 95%. The standard deviation of outdoor temperature is set as 5% forecasted value with the risk level $\varepsilon_2 = 5\%$. The MPC horizon $T_{mpc} = 6$. The parameters in **Algorithm 2** are: $\rho_0 = 10^{-3}, \gamma = 3, \alpha = 0.9, \varepsilon^{err} = 10^{-3}$. The detailed parameters of the test systems are provided in [36].

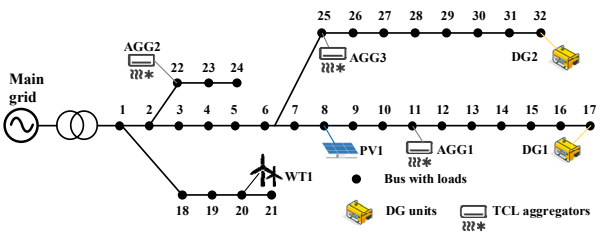

Fig. 3. The modified 33-bus distribution network.

A.1 Results of DSO Scheduling under Different Methods

The performance of different uncertainty modeling methods for DSO scheduling is compared as follows.

M1: Deterministic optimization with on predicted net loads.

M2: ICCP with polytope uncertainty set (39).

M3: DRJCCP based on Bonferroni approximation and moment information [10].

M4: DRJCCP based on data-driven Wasserstein distance [12].

M5: The statistically feasible JCCP with uncertainty set (28).

M6: The statistically feasible JCCP with uncertainty set reconstruction [14], [15].

M7: The statistically feasible JCCP with the optimal polytope uncertainty set estimated by (40).

M8: The statistically feasible JCCP with the proposed OSA.

TABLE I
Performance under Different Methods in IEEE 33 Bus System

	DSO cost	EPR	PRP	Violation rate	Solving time
M1	\$10706	-	-	99.8%	1.27s
M2	\$12090	12.9%	17.3%	75.3%	1.84s
M3	\$15501	44.8%	60.3%	0.5%	1.54s
M4	\$13610	27.1%	39.1%	4.5%	114.65s
M5	\$16496	54.1%	69.2%	0.3%	1.71s
M6	\$13975	30.5%	40.4%	4.0%	3.25s
M7	\$13594	27.0%	38.5%	4.6%	1.69s
M8	\$13517	26.3%	37.8%	4.8%	3.41s

The performance of different methods is summarized in Table I. The economic regret percentage (ERP) is defined as the additional cost (as a percentage) relative to **M1** and power regret percentage (PRP) as the additional scheduled power (as a percentage) relative to expected values. The smaller ERP and PRP imply the better scheduling decisions under the given risk level. Firstly, as expected, **M1** account for no uncertainties and **M2** ignores the correlation between constraints, leading to high violation rate. **M3** meets the required violation rate but it obtains the over-conservative dispatching solutions because it only uses the information of moment. **M4** achieves a good dispatching results by adjusting the Wasserstein radius. However, the computation speed is low due to the introduction of massive dual variables related to the sample size.

As for the statistically feasible JCCP based **M5-M8**, the modeling of the uncertainty set has a significant influence on the scheduling economy of the DSO. Firstly, the model of **M5** shows the highest cost which implies the importance of leveraging the specific problem structure. **M6** improves the results of **M5** by reconstructing the polytope uncertainty set. However, the uncertainty set shape of **M6** determined by **M5** is suboptimal, which shows a higher cost (3.4% more) than **M8**. **M7** determines the shape of uncertainty set based on (40), which results in a worse economic performance than **M9**.

M9 utilizes the proposed OSA method to optimize the shape of uncertainty set, achieving the best economic performance among all methods as well as satisfying the violation probability requirements. The computation efficiency is also high for the day-ahead optimization of DSO.

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A.2 Results of TCLs under Different Aggregation Methods

To validate the superiority of the proposed MPC-based aggregation method, we compare the proposed dynamic aggregation method with existing static aggregation methods. Additionally, since the reinforcement learning (RL) method can be integrated in the proposed dynamic aggregation framework, we also test the performance of RL method. The on-policy proximal policy optimization (PPO) is adopted for TCL aggregators to learn the scheduling strategy due to its robust performance and few parameters to tune [37]. The settings and training process of RL method are provided in [36].

Method 1: The proposed aggregation method (affine scaling matrix + MPC rolling optimization).

Method 2: The proposed aggregation method (affine scaling matrix + RL rolling optimization).

Method 3: The proposed aggregation method (uniform scaling factor + MPC rolling optimization).

Method 4: The polytope-based inner-approximation method with affine scaling matrix [22].

Method 5: The polytope-based inner-approximation method where the scaling factor is same in all dimensions [20].

Method 6: The polytope-based outer-approximation method.

We calculate the total aggregated flexible region for three TCL aggregators with total 270 TCLs. The average scaling factor (ASF) equals $\sum_i \left|\mathrm{tr}\left(\boldsymbol{B}_i\right)\right| / T$ in Method 1,2 and 4, $\sum_{i,t} \beta_{i,t} / T$ in Method 3, and $\sum_i \beta_i$ in Method 5 and 6. A larger scaling factor indicates a higher flexible capacity of TCLs. The operational results are summarized in Table II and the illustration of power bounds is shown in Fig. 4.

TABLE II
COMPARATIVE RESULTS UNDER DIFFERENT AGGREGATION METHODS

Method	1	2	3	4	5	6
ASF	221.8	211.4	206.6	198.1	181.2	276.3
Total cost ($)	519.5	530.1	533.5	553.3	579.1	391.7
Discomfort hours	0	0	0	0	0	8h
Calculating time (s)	90.3	78.8	82.2	66.5	60.1	55.7

It is observed that Method 6 has the lowest total cost with the largest scaling factor. However, this method yields infeasible operation region and leads to 8 hours thermal discomfort of users. In contrast, the inner approximation methods of Method 1-5 ensure the operation constraints of TCLs. Furthermore, different inner approximation methods are compared. The total costs of TCL aggregators are reduced by 11.3%, 9.2%, 8.5% and 4.5% and scaling factors increase by 22.2%, 16.7%, 13.9% and 9.3% in Method 1-4 with the aid of affine transformation and dynamic aggregation compared to Method 5. The affine transformation can better approximate the high dimensional feasible region of TCLs and MPC/RL method can dynamically update $\boldsymbol{H}_k^{base}, \boldsymbol{h}_k^{base}$ based on the current states of TCLs in the rolling horizon. Additionally, MPC outperforms the RL method in cost reduction while needs more time to calculate the aggregation factors and scheduling decisions. This is because the MPC method is more capable of solving linear models of TCL aggregators while the RL method is faster in making decisions based on data-driven model.

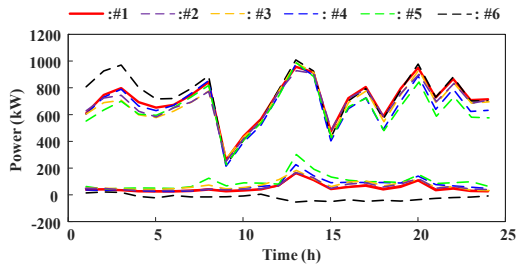

Fig. 4. The power bounds of aggregated TCLs under different methods.

Finally, the calculating time of the proposed method is larger than other methods without MPC since the proposed method needs to update the aggregation factors and scheduling decisions iteratively. Nevertheless, it is acceptable in the day-ahead optimization, and the influences of MPC parameter T_{mpc} on computational efficiency and aggregation accuracy will be further analyzed in the next subsection.

A.3 Sensitivity Analysis of MPC Horizon T_{mpc}

Due to the iterative updating of prototype's parameters, more computational time is needed to solve the MPC-based aggregation. To alleviate this issue, MPC horizon should be designed carefully to make a trade-off between aggregation performance and computational complexity. We implement the proposed algorithm with different T_{mpc} to test its performance under 270 TCLs and the results are presented in Fig. 5.

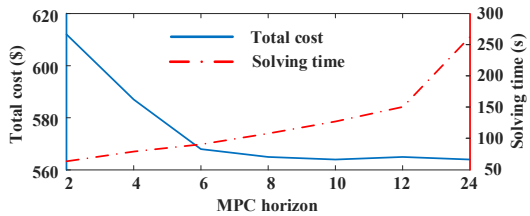

Fig. 5. Sensitive analysis of MPC horizon.

Observing Fig. 5, the computational time increases linearly with the MPC horizon due to the higher dimension of $\boldsymbol{B}_{k,i}, \boldsymbol{t}_{k,i}$ in the TCL inner approximation problem. Secondly, the total cost reduces linearly when $T_{mpc} \leq 6$ since a larger T_{mpc} helps aggregators better optimize holistic dispatching results. However, the total cost curve changes slightly when $T_{mpc} \in [6, 24]$. This indicates that the excessive MPC horizon does not improve the economical operation of aggregators while leads to lower computational efficiency. Thereby, $T_{mpc} = 6$ is considered as the best choice in this case.

A.4 Results of Proposed Asynchronous R-ADMM Method

The comparative results under the proposed R-ADMM and classical ADMM with and without increasing ρ are illustrated in Fig. 6. The proposed algorithm terminates in 7 iterations and the total computation time is less than 240s, which is faster than classical ADMM with 9 iterations. Meanwhile, the increasing ρ can accelerate the iteration process significantly compared to the constant ρ (with 38 iterations) as shown in Fig. 6.

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Besides, the total cost of all participants solved by R-ADMM is \$14037, which is almost the same as that of centralized method (\$14019).

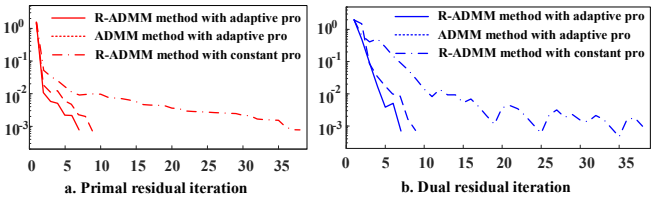

Fig. 6. Iteration process under different distributed algorithms.

To test the robustness of **Algorithm 2**, $n_d^{\max}$ is defined as the rounds of packet losses and set to 5 rounds. Besides, we assume that the first element of C_k is 0 and the difference between any two adjacent elements follows the discrete uniform distribution within $\left[1, n_d^{\max}\right]$. The convergence results of the proposed method are shown in Fig. 7. Firstly, the optimal costs of DSO and AGGs under communication failure are 13520, 198, 171 and 150 dollars, respectively. The maximum error rate of **Algorithm 2** is 0.2% deviated from the centralized results, which achieves a comparable solution accuracy even if the communication failure exists. Secondly, the iterations increase to 34, which indicates that asynchrony slows down the convergence of the proposed algorithm. Nevertheless, the proposed model is still feasible under communication failure.

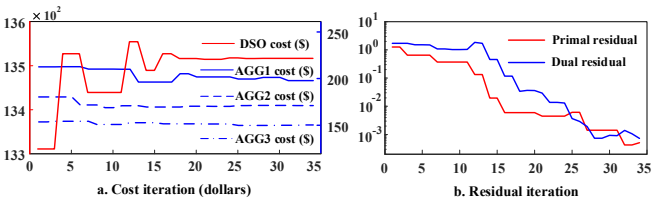

Fig. 7.Convergence of the proposed algorithm under communication failure.

B. Modified IEEE 123-Bus System Test

B.1 Performance Evaluation under IEEE 123-Bus System

The scalability of the proposed model and method is evaluated through a modified IEEE 123-bus ADN. 7 DGs and 16 RESs are equipped in the system. Besides, 9 TCL aggregators with total 4500 TCLs are integrated to operate with the DSO. The data and topology of the test system are given in [36]. The comparative results under different methods are summarized in Table III, where the uncoordinated method is the hysteretic ON/OFF local control for TCLs [31].

TABLE III
RESULTS UNDER DIFFERENT METHODS IN IEEE 123 BUS SYSTEM

Method	R-ADMM	Centralized	Uncoordinated
The cost of DSO	19968	19941	23001
Total cost of TCLs	11150	11162	12747
Total system cost	31118	31103	35748
Solving time	1472.1s	1294.3s	4.9s

Firstly, the model under the uncoordinated method has the largest costs since TCLs consume electricity without considering energy price and provides no ancillary service to earn profits. Secondly, the costs of different participants under R-ADMM method are almost the same as those in a centralized way. Finally, the solving time of R-ADMM in the parallel computing way is 1472s within 12 iterations in average, which is applicable to the large-scale system in the day-ahead scheduling. The detailed scheduling results are given in [36].

B.2 Discussion on the Implementation on Real Systems

The computation complexity of the proposed method may depend on the number of TCLs in the aggregation process, the number of aggregators engaged in the coordination process, and the software and hardware conditions for algorithm execution.

Several potential acceleration strategies merit emphasis when applying the algorithm to real-world large-scale systems. First, the calculation of $\boldsymbol{B}_{k,i}, \boldsymbol{t}_{k,i}$ can be distributed across individual TCLs via edge computing, which aims to accelerate the aggregation process. Second, a larger error tolerance for R-ADMM can be adopted for efficiency-focused operators to achieve a faster convergence process while keeping the solution error within permissible limits. Finally, enhancing the communication structure and hardware for algorithm execution will facilitate its practical implementation in real systems.

VI. CONCLUSION

In this paper, an asynchronous distributed model between the DSO and TCL aggregators is proposed. The uncertainties of nodal net loads and outdoor temperature are modeled by JCCs and the tractable reformulation by statistically feasible RO is derived with the optimal uncertainty set construction. Moreover, a MPC-based aggregation method is designed to improve the aggregated flexible capacities of TCLs. Finally, a R-ADMM based distributed method is proposed to accelerate the convergence as well as hedging against communication failures. The effectiveness of the proposed model and methods is verified via case studies in IEEE 33-bus and 123-bus systems. The key findings are summarized as follows:

1) The proposed statistically feasible RO with OSA for uncertainty set construction reduces 3.4% cost compared to self-improving reconstruction method as well as achieving high computational efficiency.

2) The proposed aggregation method helps TCL aggregators reduce by 4.5% and 5.9% operation costs as well as the scaling factor increasing by 9.3% and 10.3% with the aid of affine transformation and MPC rolling optimization.

3) The proposed asynchronous R-ADMM method can accelerate the iteration process compared to classical ADMM and enhance the robustness against communication failures.

For future research, the authors intend to extend the model to explicitly integrate aggregator-level power quality (PQ) operating regions. A specific implementation could involve embedding a voltage-informed temporal PQ design of experiments (DOE), which would serve to enhance reactive power support capabilities. Furthermore, investigating the uncertain social behaviors exhibited by consumers and quantifying how these behaviors affect the flexible capacities of TCLs represents a promising and meaningful direction.

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APPENDIX

A. Linearized Three-phase Branch Flow Model

The linearized three-phase branch flow model developed in [38] is adopted as part of the power flow constraints for the unbalanced distribution network, which is widely used for energy trading [39] and voltage control [40]. It specifies that the phase angle difference of the sequential phases *a, b, c* is 2/3π. Each TCL aggregator is connected to a particular bus via a single-phase line, which is also known as the tie-line.

First, the impedance $\boldsymbol{Z}_{ij}$ of the power line from bus *i* to bus *j*, the squared bus voltage magnitudes $\boldsymbol{v}_{j,t}$, active and reactive power flows $\boldsymbol{P}_{ij,t}$ and $\boldsymbol{Q}_{ij,t}$ from bus *i* to bus *j*, tie-line active and reactive power $\boldsymbol{p}_{j,t}^{tie}$ and $\boldsymbol{q}_{j,t}^{tie}$, active aggregated TCL power $\boldsymbol{p}_{j,t}^{agg}$ are represented as

$$\boldsymbol{Z}_{ij}=\begin{bmatrix} z_{ij}^{aa} & z_{ij}^{ab} & z_{ij}^{ac} \\ z_{ij}^{ba} & z_{ij}^{bb} & z_{ij}^{bc} \\ z_{ij}^{ca} & z_{ij}^{cb} & z_{ij}^{cc} \end{bmatrix}=\begin{bmatrix} r_{ij}^{aa} & r_{ij}^{ab} & r_{ij}^{ac} \\ r_{ij}^{ba} & r_{ij}^{bb} & r_{ij}^{bc} \\ r_{ij}^{ca} & r_{ij}^{cb} & r_{ij}^{cc} \end{bmatrix}+j\begin{bmatrix} x_{ij}^{aa} & x_{ij}^{ab} & x_{ij}^{ac} \\ x_{ij}^{ba} & x_{ij}^{bb} & x_{ij}^{bc} \\ x_{ij}^{ca} & x_{ij}^{cb} & x_{ij}^{cc} \end{bmatrix}$$
$$=\boldsymbol{R}_{ij}+j\boldsymbol{X}_{ij} \tag{A.1}$$

$$\boldsymbol{v}_{j,t}=\left[v_{j,t}^{a},v_{j,t}^{b},v_{j,t}^{c}\right]^{T} \tag{A.2}$$

$$\boldsymbol{P}_{ij,t}=\left[P_{ij,t}^{a},P_{ij,t}^{b},P_{ij,t}^{c}\right]^{T} \tag{A.3}$$

$$\boldsymbol{Q}_{ij,t}=\left[Q_{ij,t}^{a},Q_{ij,t}^{b},Q_{ij,t}^{c}\right]^{T} \tag{A.4}$$

$$\boldsymbol{p}_{j,t}^{tie}=\left[p_{j,t}^{tie,a},p_{j,t}^{tie,b},p_{j,t}^{tie,c}\right]^{T} \tag{A.5}$$

$$\boldsymbol{q}_{j,t}^{tie}=\left[q_{j,t}^{tie,a},q_{j,t}^{tie,b},q_{j,t}^{tie,c}\right]^{T} \tag{A.6}$$

$$\boldsymbol{p}_{j,t}^{agg}=\left[p_{k,j,t}^{agg,a},p_{k',j,t}^{agg,b},p_{k'',j,t}^{agg,c}\right]^{T} \tag{A.7}$$

where $\boldsymbol{R}_{ij}$ and $\boldsymbol{X}_{ij}$ are the three-phase resistance and reactance between node *i* and *j*. $p/q_{j,t}^{tie,\varphi};\varphi\in\{a,b,c\}$ denotes the tie-line injection power except for TCLs at node *j* phase *φ*, which equals $p_{g,j,t}^{G,\varphi}-p_{j,t}^{net,\varphi}$ and $q_{g,j,t}^{G,\varphi}-q_{j,t}^{net,\varphi}$. $p_{k,j,t}^{agg,\varphi}$ is the active power of *k*-th TCL aggregator. Meanwhile, for the tie-line with fewer phases, the bus generation and load related to the missing phases are set to zero. Then, the three-phase power flow can be formulated as:

$$\boldsymbol{p}_{j,t}^{tie}-\boldsymbol{p}_{j,t}^{agg}+\boldsymbol{P}_{ij,t}=\sum_{k:j\to k}\boldsymbol{P}_{jk,t} \tag{A.8}$$

$$\boldsymbol{q}_{j,t}^{tie}+\boldsymbol{Q}_{ij,t}=\sum_{k:j\to k}\boldsymbol{Q}_{jk,t} \tag{A.9}$$

$$\boldsymbol{v}_{j,t}=\boldsymbol{v}_{i,t}-2\left(\tilde{\boldsymbol{R}}_{ij}\boldsymbol{P}_{ij,t}+\tilde{\boldsymbol{X}}_{ij}\boldsymbol{Q}_{ij,t}\right) \tag{A.10}$$

where

$$\tilde{\boldsymbol{Z}}_{ij}=\tilde{\boldsymbol{R}}_{ij}+j\tilde{\boldsymbol{X}}_{ij}=Re\left(\boldsymbol{a}\boldsymbol{a}^{H}\right)\odot\boldsymbol{R}_{ij}+Im\left(\boldsymbol{a}\boldsymbol{a}^{H}\right)\odot\boldsymbol{X}_{ij}$$
$$+j\left[-Im\left(\boldsymbol{a}\boldsymbol{a}^{H}\right)\odot\boldsymbol{R}_{ij}+Re\left(\boldsymbol{a}\boldsymbol{a}^{H}\right)\odot\boldsymbol{X}_{ij}\right]$$
$$\boldsymbol{a}=\left[1,e^{-j2\pi/3},e^{-j2\pi/3}\right]^{T}$$
$$\boldsymbol{a}^{\mathrm{H}}\ \text{Hermitian transpose of}\ \boldsymbol{a}$$
$$\odot\ \text{Element - wise multiplization}.$$

In addition, the constraints for the injection active/reactive power and the square of the three-phase bus voltage magnitude are given by

$$\underline{p}_{j}^{tie}\le\boldsymbol{p}_{j,t}^{tie}-\boldsymbol{p}_{j,t}^{agg}\le\overline{p}_{j}^{tie};\forall j\in\mathcal{N},\forall t\in[T] \tag{A.11}$$

$$\underline{q}_{j}^{tie}\le\boldsymbol{q}_{j,t}^{tie}\le\overline{q}_{j}^{tie} \tag{A.12}$$

$$\underline{v}_{j}\le\boldsymbol{v}_{j,t}\le\overline{v}_{j} \tag{A.13}$$

B. Results under Unbalanced IEEE 33-Bus system

To verify the applicability of the proposed methods in the unbalanced distribution networks, we adopt a well-designed unbalanced IEEE 33-bus system established in [41]. The configuration is similar to that in Fig. 3 and TCL aggregator 1-3 are connected to phase A, B, C at bus 11, 22, 25, respectively. The results of different uncertainty modeling methods for DSO scheduling are shown in Table A.I.

TABLE A.I
RESULTS IN UNBALANCED IEEE 33 BUS SYSTEM

	DSO cost	*Delta cost	Violation rate	Solving time
M1	\$10793	\$87	99.7%	1.04s
M2	\$12162	\$72	75.5%	1.51s
M3	\$15582	\$81	0.4%	1.27s
M4	\$13685	\$75	4.6%	110.3
M5	\$16577	\$81	0.5%	1.41s
M6	\$14061	\$86	4.0%	2.97s
M7	\$13666	\$72	4.7%	1.39s
M8	\$13591	\$74	4.8%	2.80s

*Delta cost: the cost difference between unbalance case and balanced case

The results in the unbalanced case are similar to those in the balanced case with little cost increase due to the unbalanced system, which indicates the general applicability of the proposed methods. Besides, the three-phase voltage profiles at the peak load time are illustrated in Fig.A1 and the detailed operation results are provided in [36]. It is clear that the voltage magnitudes at all nodes remain in the range of ±2% indicating the safe operation of the distribution network.

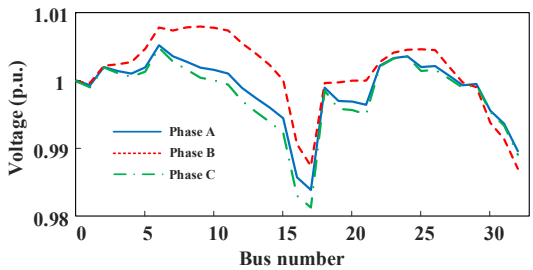

Fig. A1. Voltage profiles in unbalanced system at the peak load time.

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This article has been accepted for publication in IEEE Transactions on Power Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPWRS.2026.3670270

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This article has been accepted for publication in IEEE Transactions on Power Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPWRS.2026.3670270

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